Caffeine intake and its association with urinary incontinence in United States men: results from National Health and Nutrition Examination Surveys ...

PURPOSE: Epidemiological studies in women have revealed an association between caffeine intake and urinary incontinence, although evidence among men is limited. Therefore, we evaluated the association between caffeine intake and urinary incontinence in United States men. MATERIALS AND METHODS: Data were used from male NHANES (National Health and Nutrition Examination Surveys) 2005-2006 and 2007-2008 participants. Urinary incontinence was defined using a standard questionnaire with Incontinence Severity Index scores 3 or greater categorized as moderate to severe. Structured dietary recall was used to determine caffeine consumption (mg per day), water intake (gm per day) and total dietary moisture (gm per day). Stepwise multivariable logistic regression models were used to assess the association between caffeine intake at or above the 75th and 90th percentiles and moderate to severe urinary incontinence, controlling for potential confounders, urinary incontinence risk factors and prostate conditions in men age 40 years or older. RESULTS: Of the 5,297 men 3,960 (75%) were 20 years old or older with complete data. Among these men the prevalence of any urinary incontinence was 12.9% and moderate to severe urinary incontinence was 4.4%. Mean caffeine intake was 169 mg per day. Caffeine intake at the upper 75th percentile (234 mg or more daily) and 90th percentile (392 mg or more per day) was significantly associated with having moderate to severe urinary incontinence (1.72, 95% 1.18-2.49 and 2.08, 95% 1.15-3.77, respectively). In addition, after adjusting for prostate conditions, the effect size for the association between caffeine intake and moderate to severe urinary incontinence remained. CONCLUSIONS: Caffeine consumption equivalent to approximately 2 cups of coffee daily (250 mg) is significantly associated with moderate to severe urinary incontinence in United States men. Our findings support the further study of caffeine modification in men with urinary incontinence.